Class Disparities and Discrimination in Traffic Stops and Searches

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Motivation

- Class may shape how police engage with civilians (Robison, 1936)
- Low-SES neighborhoods are policed more aggressively (Fagan et al., 2010; MacDonald, 2021; Chen et al., 2025)
- Q: in the same context, are low-SES civilians policed more aggressively?
- Potential implications for:
  - Mobility and inequality
  - Trust in criminal justice institutions
  - Effectiveness of policing

This research

- We study class disparities in searches during traffic stops
- Use detailed data on stops conducted by Texas Highway Patrol
- Measure class differences in search rates and contraband yield
- Develop and implement test for whether troopers engage in class *discrimination*
- Consider different explanations for class disparities

Preview of findings

- Class disparities in search rates are *large*
- Searches of low-income motorists are *less* likely to yield contraband
- Class disparities at least in part reflect class *discrimination*
  - Same motorist is more likely to be searched in a low-status vehicle and is stopped by more search-intensive troopers

Literature context

- Research on profiling that has documented racial disparities in
  - Vehicle stops: Grogger and Ridgeway (2006); Pierson et al. (2020)
  - Searches: Knowles et al. (2001); Anwar and Fang (2006); Close and Mason (2007); Antonovics and Knight (2009); Marx (2022); Feigenberg and Miller (2022)
  - Tickets: Goncalves and Mello (2021)
- We know much less about the role of class, probably due to lack of data

Talk Outline

1. Institutional context and data
2. Search and hit rates by income
3. Test for trooper discrimination

Institutional context

- We study traffic stops conducted by Texas Highway Patrol
- Most stops occur on state and interstate highways
- Trooper can:
  - Stop motorists if they observe or have reasonable suspicion of a violation
  - During stop, investigate if motorist is carrying contraband, including illicit drugs and weapons
  - Conduct a search if they have probable cause to believe a law has been broken
  - Ask for and receive motorist consent to conduct a search

Data

- 16 million vehicle stops conducted by Texas Highway Patrol (2009-2015)
  - Information on stop time and location, vehicle characteristics, motorist race and gender, stop and search outcomes, trooper ID
  - Unique feature: includes motorist’s full name and address
    - Match multiple traffic stops to the same motorist
- Individual-level criminal histories from Texas Computerized Criminal History System
  - Measure arrest and court outcomes associated with stop
- Commercial address history data (Infogroup/Data Axle) used to facilitate matching traffic stops and criminal history to a given motorist

Measuring household income

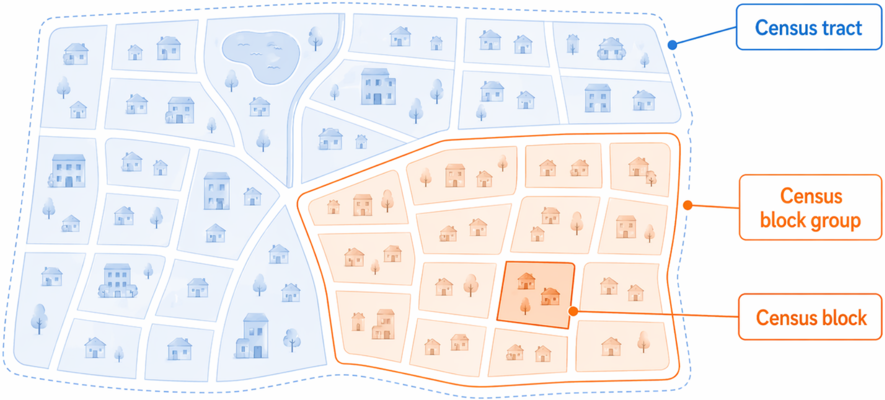
- 2009-2013 ACS block group-level income data and ATTOM property assessment data used to infer household income
  - Single-family residential properties: assign to p^{th} percentile of homeowner household income distribution if property value falls in p^{th} percentile
  - For all others (multifamily housing, apartment complexes, etc.): assign median household income category among renters
  - Finally, allocate across 16 income intervals available for pooled block group sample
  - Results insensitive to alternative approaches, including assigning all motorists to block group median income

1. Start with location

We observe the exact address of the motorist.

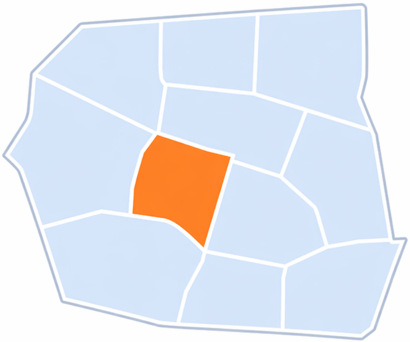
2. Define a Census block group

A Census block group is a cluster of nearby Census blocks within a Census tract.



3. Census gives the local income distribution

For each block group, Census reports how many households fall in each income range.



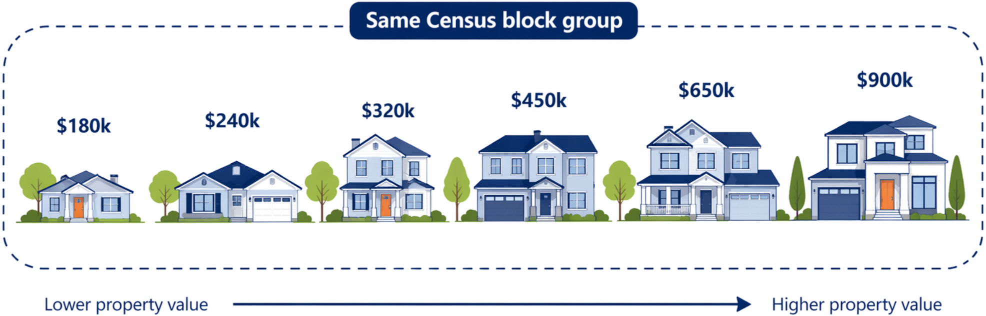
Example block group
(within a Census tract)

Households by income range

<$25k	**210**	
$25k–$50k	**380**	
$50k–$100k	**540**	
$100k–$200k	**310**	
$200k+	**160**	

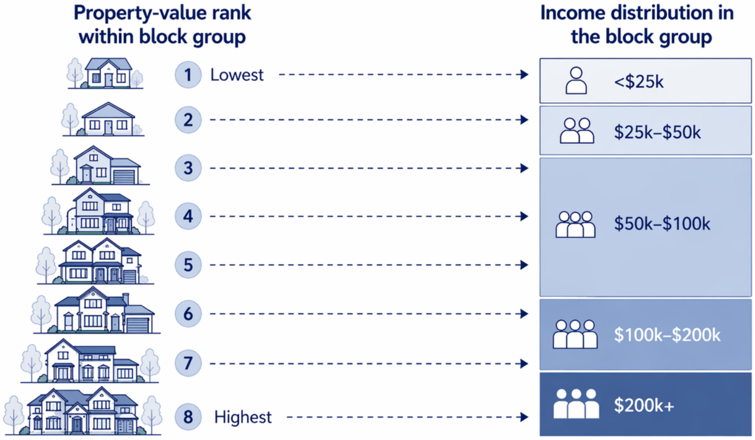
4. Homes in the block group have different values

Property values vary within the same block group.



5. Infer household income by ranking homes

Within a block group, we assign households to the local income distribution based on property-value rank.



This yields an estimate of household income, not a direct measure.

Figure: Distribution of household income across stops

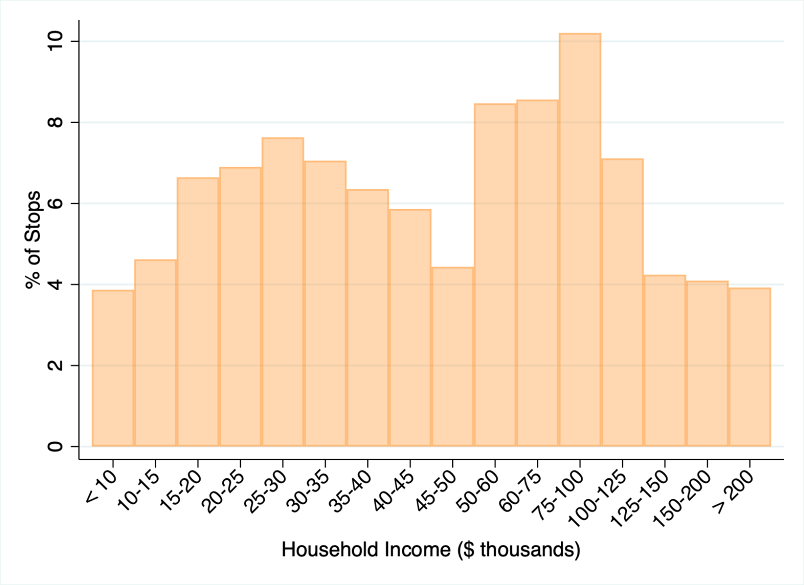


Table: Traffic stop descriptive statistics

	All Stops			All Searches		
	Below Median	Above Median	All	Below Median	Above Median	All
Black	10.11	8.64	9.42	16.80	15.04	16.18
Hispanic	37.72	24.64	31.61	39.39	29.72	36.01
White	49.85	63.21	56.09	42.00	52.62	45.71
Female	35.08	34.53	34.83	19.81	18.96	19.51
Log household income	9.94	11.34	10.59	9.91	11.23	10.37
	(0.61)	(0.49)	(0.89)	(0.61)	(0.45)	(0.84)
Search rate	2.34	1.44	1.92	100	100	100
Unconditional hit rate	0.82	0.56	0.70	34.45	38.55	35.89
Implied percent pretextual	11.15	8.87	10.08	20.51	19.03	19.99
	(14.92)	(13.83)	(14.47)	(15.83)	(15.94)	(15.88)
Moving	68.48	74.39	71.24	60.28	62.75	61.14
Driving while intoxicated	2.26	1.33	1.83	22.13	21.61	21.95
Speeding	55.44	63.54	59.22	28.27	32.96	29.91
Equipment	21.28	15.98	18.81	18.74	17.08	18.16
Regulatory	34.52	28.53	31.72	36.14	30.58	34.20
Prior felony arrests	0.145	0.0818	0.115	0.570	0.472	0.536
	(0.771)	(0.564)	(0.683)	(1.554)	(1.403)	(1.503)
Prior misdemeanor arrests	0.346	0.219	0.286	1.265	1.130	1.218
	(1.370)	(1.041)	(1.229)	(2.708)	(2.469)	(2.628)
Observations	5,868,369	5,138,516	11,006,885	137,513	74,020	211,533

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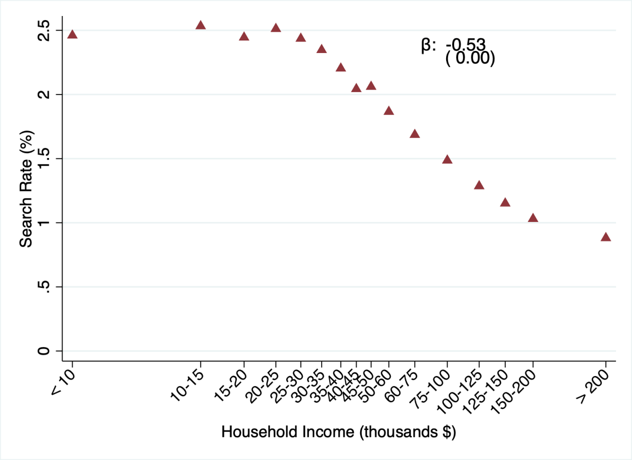
Figure: Search rates are decreasing in motorist income



- Motorists in top (bottom) income quintile searched in 1.1% (2.5%) of stops; for comparison, Black and Hispanic motorists are searched 150% and 60% more often than White motorists

Do class disparities reflect other contextual differences?

- These disparities may reflect differences in stop context, including the location and time, or other motorist demographics
  - We estimate regression models of the form:

$$Y_{it} = \alpha_{\ell_{i,t}\tau(t)y(t)} + \beta \text{log(income)}_{it} + X_{it}\Gamma + \phi_{p(it)} + \epsilon_{it} \quad (1)$$

  - $\alpha_{\ell_{i,t}\tau(t)y(t)}$ are fixed effects for combination of trooper patrol area ("sergeant area"), time of week (quarter of day, weekday or weekend), and year
  - X_{it} is a vector of motorist demographic characteristics, including race, gender, and criminal history
  - $\phi_{p(it)}$ are trooper fixed effects
- Could also reflect differences in violations associated with stops
  - Qualitatively similar findings if we limit to speeding stops

Table: Search rates by motorist income

Outcome:	Search ($\times$100)				
	(1)	(2)	(3)	(4)	(5)
log Household Income	-0.53 (0.00)	-0.54 (0.00)	-0.48 (0.00)	-0.38 (0.00)	-0.35 (0.00)
Sgt. Area $\times$ Time of Week $\times$ Year FEs		✓	✓	✓	✓
Motorist Demographics			✓	✓	✓
Motorist Criminal History				✓	✓
Trooper FE					✓
Mean of DV	1.92				
Observations	11,022,012				

How do hit rates vary with motorist income?

- Low-income motorists are more likely to be searched
- One potential explanation: low-income motorists are more likely to carry contraband
- We measure contraband yield (“hit rates”) by motorist income
  - No evidence of differences by income in coarse contraband type

Figure: Hit rates Are increasing in motorist income

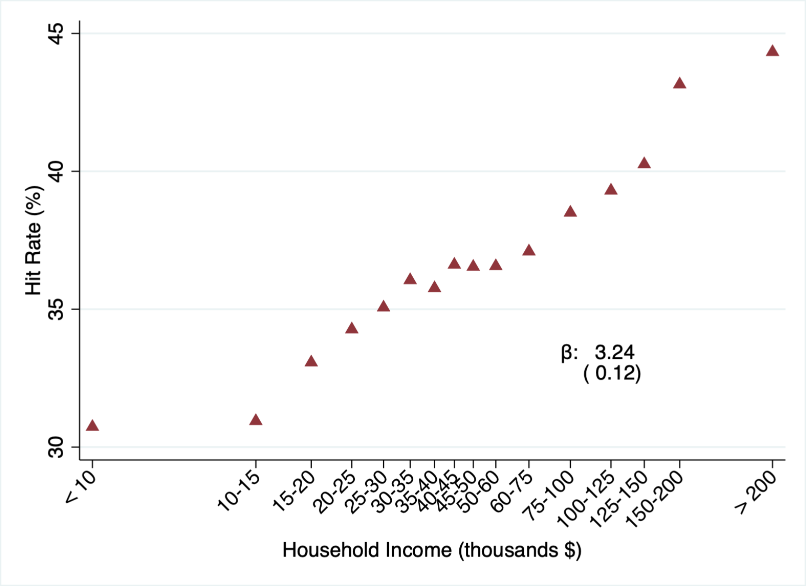


Table: Hit rates by motorist income

Outcome:	Contraband Recovery ($\times$100)				
	(1)	(2)	(3)	(4)	(5)
log Household Income	3.25	1.56	1.36	1.39	1.30
	(0.12)	(0.12)	(0.12)	(0.12)	(0.12)
Sgt. Area $\times$ Year FEs		✓	✓	✓	✓
Motorist Demographics			✓	✓	✓
Motorist Criminal History				✓	✓
Trooper FE					✓
Mean of DV			35.88		
Observations			211,546		

Talk Outline

1. Institutional context and data
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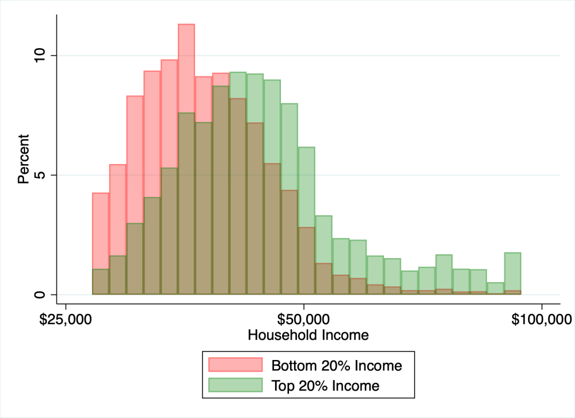
3. **Test for trooper discrimination**

Is this class discrimination?

- Troopers may *discriminate* on the basis of perceived class; or they may target searches based on other motorist characteristics that are *correlated* with perceived class
- A central challenge to investigating disparities is distinguishing between the two (Charles and Guryan, 2011)
- Ideal experiment: Vary the *perceived* class of a motorist, holding behavior fixed
- Our strategy: look at the *same* motorist stopped in *different* vehicles
  - Assumption: other than motorist's perceived class, motorist characteristics (including propensity to consent to search) are fixed, or variation is uncorrelated with variation in vehicle
  - Plausible given trip to trip variation in vehicle (65% of Texas households with 1+ car own multiple vehicles, new purchases infrequent)
  - 60% of stops involve motorists that are stopped multiple times; in pairs of consecutive stops, over half involve different vehicles

Figure: Distribution of vehicle status by income



- We measure vehicle status as predicted log household income based on vehicle make, type (passenger car, pick-up truck, or SUV), and age (SD: 22 log points)

Figure: Troopers profile motorists at the search margin

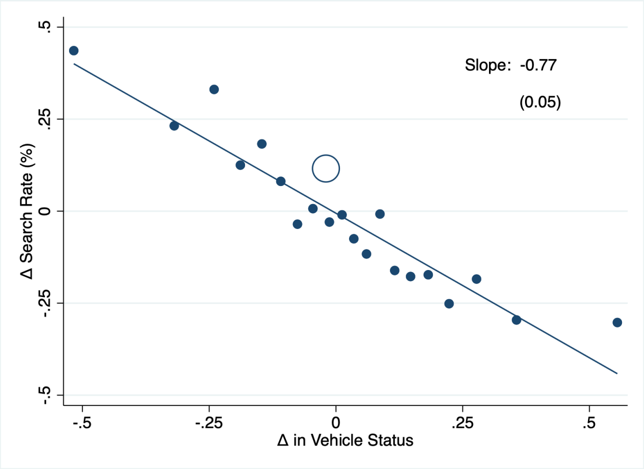
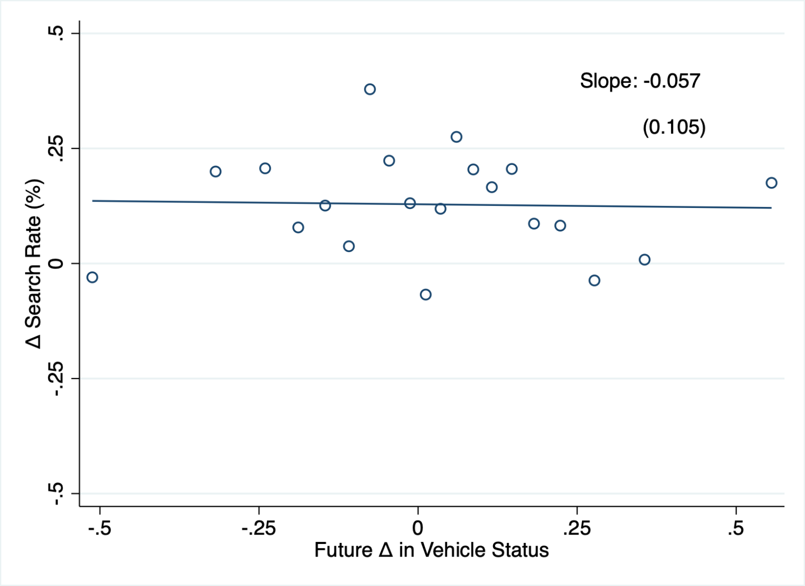


Figure: No change in search rates prior to vehicle switch



Discussion

- We document large class disparities in search rates and pretext stops
- Low-income motorists are more likely to be searched and found with contraband despite being less likely to be found with contraband conditional on being searched
- Class disparities at least in part reflect class discrimination
- Finding have important implications for fairness and equity, exposure to criminal sanctions
- We present suggestive evidence that trooper behavior is at least in part explained by anticipated downstream hassle costs

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